

PATIENT & SPECIMEN INFORMATION

Patient name	Hemingway, Ernest M.	Accession	NGS-2026-052101
Date of birth	1961-07-22 (Age 64)	Specimen	Oral swab, left lateral tongue
MRN	10345621	Collected	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	Received	2026-05-19
Clinical indication	Oral lesion — left lateral tongue	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
KRAS	NM_004985.5:c.35G>T	p.Gly12Asp	0.43%	463×	I	Pathogenic
PIK3CA	NM_006218.4:c.3140A>G	p.His1047Arg	0.70%	714×	I	Pathogenic

VARIANT ANNOTATION

KRAS p.Gly12Asp | Pathogenic (Tier I)

KRAS p.Gly12Asp (G12D) is one of the most frequently detected oncogenic driver mutations in human cancer. It locks KRAS in the GTP-bound active state, constitutively activating downstream RAS/MAPK and PI3K/AKT/mTOR proliferative signalling. In oral squamous cell carcinoma, KRAS mutations are found in 3-8% of cases and mark a subgroup with aggressive behaviour. This variant is classified Pathogenic (Tier I, COSMIC ID COSM521). The co-occurrence with PIK3CA H1047R (see below) indicates dual activation of RAS and PI3K pathways, a combination associated with resistance to single-agent targeted therapies and may influence enrolment eligibility in clinical trials. NOTE: Observed VAF (0.43%) is above the 0.2% reporting threshold; this call is supported by 2 high-quality consensus reads at 463× consensus depth. The low consensus read count (VD = 2) reflects limited target coverage at 2,000× raw input; orthogonal confirmation is recommended before clinical action.

PIK3CA p.His1047Arg | Pathogenic (Tier I)

PIK3CA p.His1047Arg is a hotspot activating mutation in the kinase domain of the p110α catalytic subunit of PI3-kinase. It constitutively activates the PI3K/AKT/mTOR pathway, promoting cell survival, proliferation, and tumour angiogenesis. PIK3CA mutations occur in 15-20% of OSCC, most commonly at E545K and H1047R hotspots. This variant is classified Pathogenic (Tier I, COSMIC ID COSM775). PIK3CA H1047R is potentially actionable: alpelisib (PI3Kα inhibitor) is FDA-approved for PIK3CA-mutated breast cancer and is under investigation in HNSCC. Multidisciplinary tumour board review for consideration of PI3K-directed therapy or clinical trial enrolment is recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, NFE2L2, NOTCH1, NSD1, PIK3R1, PTEN, TERT, TP53
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (≥400× consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 2,000× raw sequencing depth, approximately 15-20% of targets may fall below 100× consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

HOTSPOT SCREENING SUMMARY

Patient:

Hemingway, Ernest M.

Accession:

NGS-2026-052101

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	● DETECTED	P2415L	c.7244C>T	Not detected
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	● DETECTED	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)

251 targets, 77,417 bp total. Mean consensus depth: 388x. Targets below 100x consensus: 61 / 251 (24%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	163x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	27x	⚠ Below 100x threshold
DPYD	chr1:97,515,687-97,515,887	201	118x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	186x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	429x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	981x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	169x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	726x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	181x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	215x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	152x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	93x	⚠ Below 100x threshold
PTEN	chr10:87,965,267-87,965,489	223	463x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	312x	✓ Covered
HRAS	chr11:533,433-533,632	200	39x	⚠ Below 100x threshold
HRAS	chr11:533,746-533,964	219	247x	✓ Covered
HRAS	chr11:534,192-534,342	151	178x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	110x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	709x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	56x	⚠ Below 100x threshold
KRAS	chr12:25,245,254-25,245,404	151	1,113x	● VARIANT DETECTED
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	137x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	680x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	297x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	392x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	1,406x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	35x	⚠ Below 100x threshold
KMT2D	chr12:49,026,162-49,027,342	1,181	1,993x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	1,007x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	345x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	45x	⚠ Below 100x threshold
KMT2D	chr12:49,029,041-49,029,256	216	20x	⚠ Below 100x threshold
KMT2D	chr12:49,029,381-49,029,496	116	134x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	35x	⚠ Below 100x threshold
KMT2D	chr12:49,030,581-49,030,788	208	272x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	158x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	279x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	283x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	51x	⚠ Below 100x threshold
KMT2D	chr12:49,034,562-49,034,686	125	488x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	107x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	677x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	535x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	325x	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	55x	⚠ Below 100x threshold
KMT2D	chr12:49,041,635–49,041,725	91	1,879x	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	38x	⚠ Below 100x threshold
KMT2D	chr12:49,042,069–49,042,350	282	249x	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	157x	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	58x	⚠ Below 100x threshold
KMT2D	chr12:49,043,056–49,043,206	151	434x	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	712x	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	140x	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	994x	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	646x	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	319x	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	116x	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	81x	⚠ Below 100x threshold
KMT2D	chr12:49,046,045–49,046,194	150	282x	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	472x	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	273x	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	62x	⚠ Below 100x threshold
KMT2D	chr12:49,048,639–49,048,789	151	70x	⚠ Below 100x threshold
KMT2D	chr12:49,049,085–49,049,238	154	259x	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	454x	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	577x	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	295x	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	45x	⚠ Below 100x threshold
KMT2D	chr12:49,053,187–49,053,341	155	468x	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	35x	⚠ Below 100x threshold
KMT2D	chr12:49,053,958–49,054,160	203	150x	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	14x	⚠ Below 100x threshold
KMT2D	chr12:49,054,508–49,054,771	264	93x	⚠ Below 100x threshold
KMT2D	chr12:49,054,880–49,055,046	167	1,501x	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	246x	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	158x	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	176x	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	460x	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	181x	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	219x	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	145x	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	1,867x	✓ Covered
AJUBA	chr14:22,981,241–22,982,286	1,046	137x	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	368x	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	107x	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	207x	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	30x	⚠ Below 100x threshold
TP53	chr17:7,674,161–7,674,310	150	584x	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	126x	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	86x	⚠ Below 100x threshold
TP53	chr17:7,675,974–7,676,292	319	1,358x	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	173x	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	274x	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	51x	⚠ Below 100x threshold
KEAP1	chr19:10,489,172–10,489,388	217	1,616x	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	833x	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	1,616x	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	178x	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	231x	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	187x	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	69x	⚠ Below 100x threshold
CASP8	chr2:201,272,618–201,272,796	179	101x	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	78x	⚠ Below 100x threshold
CASP8	chr2:201,274,869–201,274,973	105	55x	⚠ Below 100x threshold
CASP8	chr2:201,276,807–201,276,988	182	250x	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	61x	⚠ Below 100x threshold
CASP8	chr2:201,286,439–201,286,611	173	139x	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	978x	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	61x	⚠ Below 100x threshold
CUL3	chr2:224,481,872–224,482,098	227	60x	⚠ Below 100x threshold
CUL3	chr2:224,495,812–224,495,986	175	179x	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	992x	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	432x	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	111x	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	244x	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	17x	⚠ Below 100x threshold
CUL3	chr2:224,506,838–224,507,023	186	57x	⚠ Below 100x threshold
CUL3	chr2:224,511,334–224,511,602	269	288x	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	735x	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	39x	⚠ Below 100x threshold
CUL3	chr2:224,535,508–224,535,661	154	436x	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	150x	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	35x	⚠ Below 100x threshold
EP300				
EP300	chr22:41,092,985–41,093,118	134	449x	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	322x	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	424x	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	109x	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	660x	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	1,676x	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	564x	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	196x	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	260x	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	105x	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	171x	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	413x	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	161x	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	88x	⚠ Below 100x threshold
EP300	chr22:41,151,813–41,152,032	220	408x	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	136x	✓ Covered

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Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	158×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	457×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	2,193×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	235×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	66×	⚠ Below 100× threshold
EP300	chr22:41,164,033–41,164,150	118	226×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	111×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	218×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	82×	⚠ Below 100× threshold
EP300	chr22:41,169,483–41,169,636	154	105×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	65×	⚠ Below 100× threshold
EP300	chr22:41,172,479–41,172,683	205	194×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	235×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	63×	⚠ Below 100× threshold
EP300	chr22:41,176,753–41,178,973	2,221	312×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	323×	✓ Covered
PIK3CA	chr3:179,234,074–179,234,381	308	246×	● VARIANT DETECTED
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	449×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	94×	⚠ Below 100× threshold
FAT1	chr4:186,595,669–186,595,846	178	12×	⚠ Below 100× threshold
FAT1	chr4:186,596,520–186,597,191	672	203×	✓ Covered
FAT1	chr4:186,597,662–186,597,812	151	23×	⚠ Below 100× threshold
FAT1	chr4:186,597,952–186,598,145	194	187×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	240×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	121×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	61×	⚠ Below 100× threshold
FAT1	chr4:186,603,156–186,603,997	842	58×	⚠ Below 100× threshold
FAT1	chr4:186,604,357–186,604,594	238	64×	⚠ Below 100× threshold
FAT1	chr4:186,606,050–186,606,233	184	214×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	401×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	171×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	395×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	618×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	240×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	800×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	327×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	286×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	1,146×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	199×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	283×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	857×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	124×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	535×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	230×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	388×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	731×	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	96x	⚠ Below 100x threshold
PIK3R1	chr5:68,295,400-68,295,508	109	803x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	216x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	678x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	33x	⚠ Below 100x threshold
NSD1	chr5:177,209,616-177,212,215	2,600	240x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	1,871x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	59x	⚠ Below 100x threshold
NSD1	chr5:177,239,736-177,239,885	150	204x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	213x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	1,149x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	68x	⚠ Below 100x threshold
NSD1	chr5:177,251,710-177,251,873	164	317x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	333x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	302x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	167x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	41x	⚠ Below 100x threshold
NSD1	chr5:177,273,652-177,273,804	153	148x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	1,195x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	595x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	864x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	256x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	696x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	97x	⚠ Below 100x threshold
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	45x	⚠ Below 100x threshold
EGFR	chr7:55,174,702-55,174,840	139	429x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	123x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	188x	✓ Covered
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	76x	⚠ Below 100x threshold
CDKN2A	chr9:21,970,882-21,971,228	347	73x	⚠ Below 100x threshold
CDKN2A	chr9:21,974,658-21,974,846	189	242x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	155x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	181x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	61x	⚠ Below 100x threshold
NOTCH1	chr9:136,499,092-136,499,279	188	101x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	201x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	54x	⚠ Below 100x threshold
NOTCH1	chr9:136,501,981-136,502,108	128	43x	⚠ Below 100x threshold
NOTCH1	chr9:136,502,252-136,502,508	257	563x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	512x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	747x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	2,686x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	198x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	369x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	451x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	292x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	255x	✓ Covered

FULL PANEL COVERAGE REPORT — Hemingway, Ernest M. (NGS-2026-052101)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	33x	⚠ Below 100x threshold
NOTCH1	chr9:136,509,713-136,509,981	269	1,232x	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	50x	⚠ Below 100x threshold
NOTCH1	chr9:136,511,132-136,511,291	160	76x	⚠ Below 100x threshold
NOTCH1	chr9:136,513,001-136,513,154	154	456x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	22x	⚠ Below 100x threshold
NOTCH1	chr9:136,514,490-136,514,722	233	111x	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	112x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	7,690x	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	279x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	46x	⚠ Below 100x threshold
NOTCH1	chr9:136,517,732-136,517,957	226	210x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	207x	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	619x	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	732x	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	109x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	13x	⚠ Below 100x threshold
NOTCH1	chr9:136,544,004-136,544,122	119	114x	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	1,230x	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Woolf, Virginia A.	Accession	NGS-2026-052102
Date of birth	1970-03-11 (Age 56)	Specimen	Oral swab, right buccal mucosa
MRN	20456732	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — right buccal mucosa	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
NOTCH1	NM_017617.5:c.7244C>T	p.Pro2415Leu	0.26%	769×	II	Likely Pathogenic

VARIANT ANNOTATION

NOTCH1 p.Pro2415Leu | Likely Pathogenic (Tier II)

NOTCH1 functions as a tumour suppressor in stratified squamous epithelium. The p.Pro2415Leu missense variant affects a conserved residue in the PEST domain, predicted to disrupt protein stability (SIFT: deleterious; PolyPhen-2: probably damaging). NOTCH1 loss-of-function mutations are found in 10–15% of OSCC and are associated with early-onset tumourigenesis in the oral cavity. This variant is classified Likely Pathogenic (Tier II). In the context of confirmed dysplasia, detection of a NOTCH1 tumour suppressor variant supports malignant progression risk. Clinical correlation and multidisciplinary review are recommended. NOTE: Observed VAF (0.26%) is above the 0.2% reporting threshold and is supported by 2 consensus reads at 769× consensus depth. Orthogonal confirmation is recommended before clinical action.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (≥400× consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 2,000× raw sequencing depth, approximately 15–20% of targets may fall below 100× consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:Woolf, Virginia A.

Accession:NGS-2026-052102

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	● DETECTED
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)

251 targets, 77,417 bp total. Mean consensus depth: 445x. Targets below 100x consensus: 74 / 251 (29%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	184x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	198x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	341x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	208x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	163x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	421x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	219x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	174x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	547x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	177x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	54x	⚠ Below 100x threshold
PTEN	chr10:87,960,874-87,961,138	265	501x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	200x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	549x	✓ Covered
HRAS	chr11:533,433-533,632	200	172x	✓ Covered
HRAS	chr11:533,746-533,964	219	23x	⚠ Below 100x threshold
HRAS	chr11:534,192-534,342	151	56x	⚠ Below 100x threshold
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	99x	⚠ Below 100x threshold
KRAS	chr12:25,225,594-25,225,793	200	45x	⚠ Below 100x threshold
KRAS	chr12:25,227,214-25,227,432	219	89x	⚠ Below 100x threshold
KRAS	chr12:25,245,254-25,245,404	151	574x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	57x	⚠ Below 100x threshold
KMT2D	chr12:49,022,023-49,022,171	149	3,137x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	56x	⚠ Below 100x threshold
KMT2D	chr12:49,022,570-49,022,895	326	461x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	175x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	22x	⚠ Below 100x threshold
KMT2D	chr12:49,026,162-49,027,342	1,181	208x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	243x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	294x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	131x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	18x	⚠ Below 100x threshold
KMT2D	chr12:49,029,381-49,029,496	116	205x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	122x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	190x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	76x	⚠ Below 100x threshold
KMT2D	chr12:49,031,155-49,033,984	2,830	150x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	1,497x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	174x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	65x	⚠ Below 100x threshold
KMT2D	chr12:49,034,792-49,034,955	164	75x	⚠ Below 100x threshold
KMT2D	chr12:49,037,105-49,039,009	1,905	386x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	578x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	1,972x	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	570×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	78×	⚠ Below 100× threshold
KMT2D	chr12:49,041,897–49,042,010	114	949×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	1,477×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	281×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	284×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	326×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	1,264×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	248×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	406×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	60×	⚠ Below 100× threshold
KMT2D	chr12:49,044,383–49,044,542	160	64×	⚠ Below 100× threshold
KMT2D	chr12:49,044,724–49,044,985	262	211×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	644×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	421×	✓ Covered
KMT2D	chr12:49,046,240–49,046,444	205	136×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	283×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	1,634×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	3,482×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	124×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	862×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	958×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	228×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	708×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	133×	✓ Covered
KMT2D	chr12:49,053,456–49,053,661	206	515×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	131×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	39×	⚠ Below 100× threshold
KMT2D	chr12:49,054,508–49,054,771	264	239×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	1,123×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	308×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	85×	⚠ Below 100× threshold
AJUBA	chr14:22,974,027–22,974,135	109	34×	⚠ Below 100× threshold
AJUBA	chr14:22,974,819–22,974,910	92	121×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	98×	⚠ Below 100× threshold
AJUBA	chr14:22,976,436–22,976,538	103	68×	⚠ Below 100× threshold
AJUBA	chr14:22,976,625–22,976,732	108	275×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	80×	⚠ Below 100× threshold
AJUBA	chr14:22,981,241–22,982,286	1,046	208×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	212×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	447×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	25×	⚠ Below 100× threshold
TP53	chr17:7,673,681–7,673,857	177	37×	⚠ Below 100× threshold
TP53	chr17:7,674,161–7,674,310	150	271×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	423×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	778×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	282×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	42×	⚠ Below 100× threshold
TP53	chr17:7,676,501–7,676,614	114	11×	⚠ Below 100× threshold

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	96×	⚠ Below 100× threshold
KEAP1	chr19:10,489,172–10,489,388	217	90×	⚠ Below 100× threshold
KEAP1	chr19:10,489,628–10,489,873	246	54×	⚠ Below 100× threshold
KEAP1	chr19:10,491,557–10,492,282	726	95×	⚠ Below 100× threshold
KEAP1	chr19:10,499,375–10,500,053	679	2,537×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	274×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	26×	⚠ Below 100× threshold
CASP8	chr2:201,271,496–201,271,641	146	1,120×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	102×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	251×	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	583×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	176×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	897×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	110×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	137×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	701×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	29×	⚠ Below 100× threshold
CUL3	chr2:224,495,812–224,495,986	175	19×	⚠ Below 100× threshold
CUL3	chr2:224,497,733–224,497,869	137	1,604×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	25×	⚠ Below 100× threshold
CUL3	chr2:224,502,945–224,503,092	148	89×	⚠ Below 100× threshold
CUL3	chr2:224,503,632–224,503,842	211	126×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	124×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	791×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	42×	⚠ Below 100× threshold
CUL3	chr2:224,513,504–224,513,658	155	223×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	304×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	160×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	101×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	135×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	50×	⚠ Below 100× threshold
EP300	chr22:41,117,167–41,117,841	675	44×	⚠ Below 100× threshold
EP300	chr22:41,125,844–41,126,060	217	304×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	217×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	231×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	1,919×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	323×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	328×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	241×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	190×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	270×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	58×	⚠ Below 100× threshold
EP300	chr22:41,149,018–41,149,195	178	72×	⚠ Below 100× threshold
EP300	chr22:41,149,741–41,150,218	478	61×	⚠ Below 100× threshold
EP300	chr22:41,151,813–41,152,032	220	375×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	96×	⚠ Below 100× threshold

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	436×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	78×	⚠ Below 100× threshold
EP300	chr22:41,158,392–41,158,520	129	44×	⚠ Below 100× threshold
EP300	chr22:41,160,622–41,160,742	121	290×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	54×	⚠ Below 100× threshold
EP300	chr22:41,164,033–41,164,150	118	289×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	473×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	876×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	219×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	1,864×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	349×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	1,386×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	100×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	111×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	72×	⚠ Below 100× threshold
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	64×	⚠ Below 100× threshold
PIK3CA	chr3:179,234,074–179,234,381	308	264×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	497×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	438×	✓ Covered
FAT1	chr4:186,595,669–186,595,846	178	38×	⚠ Below 100× threshold
FAT1	chr4:186,596,520–186,597,191	672	68×	⚠ Below 100× threshold
FAT1	chr4:186,597,662–186,597,812	151	208×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	54×	⚠ Below 100× threshold
FAT1	chr4:186,599,878–186,600,380	503	47×	⚠ Below 100× threshold
FAT1	chr4:186,601,249–186,601,446	198	72×	⚠ Below 100× threshold
FAT1	chr4:186,602,883–186,603,054	172	121×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	579×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	162×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	300×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	540×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	27×	⚠ Below 100× threshold
FAT1	chr4:186,611,366–186,611,795	430	47×	⚠ Below 100× threshold
FAT1	chr4:186,613,089–186,613,362	274	248×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	189×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	84×	⚠ Below 100× threshold
FAT1	chr4:186,617,688–186,621,795	4,108	68×	⚠ Below 100× threshold
FAT1	chr4:186,628,134–186,628,384	251	106×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	115×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	426×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	318×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	153×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	129×	✓ Covered
FAT1	chr4:186,663,279–186,663,633	355	875×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	63×	⚠ Below 100× threshold
TERT				
TERT	chr5:1,295,050–1,295,250	201	446×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	123×	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	87x	⚠ Below 100x threshold
PIK3R1	chr5:68,295,400-68,295,508	109	81x	⚠ Below 100x threshold
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	26x	⚠ Below 100x threshold
NSD1	chr5:177,191,864-177,192,039	176	352x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	395x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	292x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	529x	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	18x	⚠ Below 100x threshold
NSD1	chr5:177,239,736-177,239,885	150	616x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	181x	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	16,826x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	36x	⚠ Below 100x threshold
NSD1	chr5:177,251,710-177,251,873	164	5,801x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	272x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	66x	⚠ Below 100x threshold
NSD1	chr5:177,267,542-177,267,738	197	1,100x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	241x	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	2,101x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	36x	⚠ Below 100x threshold
NSD1	chr5:177,282,445-177,282,601	157	143x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	203x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	155x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	359x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	386x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	227x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	604x	✓ Covered
EGFR	chr7:55,181,273-55,181,498	226	115x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	93x	⚠ Below 100x threshold
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	467x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	187x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	653x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	99x	⚠ Below 100x threshold
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	222x	🔴 VARIANT DETECTED
NOTCH1	chr9:136,498,879-136,499,016	138	63x	⚠ Below 100x threshold
NOTCH1	chr9:136,499,092-136,499,279	188	719x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	112x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	1,403x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	138x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	61x	⚠ Below 100x threshold
NOTCH1	chr9:136,503,162-136,503,350	189	145x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	984x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	164x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	169x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	229x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	480x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	193x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	2,311x	✓ Covered

FULL PANEL COVERAGE REPORT — Woolf, Virginia A. (NGS-2026-052102)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	65x	⚠ Below 100x threshold
NOTCH1	chr9:136,509,713-136,509,981	269	181x	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	193x	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	187x	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	342x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	77x	⚠ Below 100x threshold
NOTCH1	chr9:136,514,490-136,514,722	233	111x	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	137x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	831x	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	227x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	29x	⚠ Below 100x threshold
NOTCH1	chr9:136,517,732-136,517,957	226	303x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	277x	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	29x	⚠ Below 100x threshold
NOTCH1	chr9:136,519,423-136,519,585	163	45x	⚠ Below 100x threshold
NOTCH1	chr9:136,522,830-136,523,208	379	235x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	304x	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	1,784x	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	32x	⚠ Below 100x threshold

PATIENT & SPECIMEN INFORMATION

Patient name	Dickens, Charles J.	Accession	NGS-2026-052103
Date of birth	1951-09-28 (Age 74)	Specimen	Oral swab, left floor of mouth
MRN	30567843	Collected	2026-05-19
Ordering physician	Dr. Wisdom, DDS MD	Received	2026-05-19
Clinical indication	Oral lesion — left floor of mouth	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.60%	664x	I	Pathogenic

VARIANT ANNOTATION

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His is the most frequently observed p53 mutation in human cancer. It disrupts the structural zinc-binding domain of the DNA-binding domain, abolishing wild-type tumour suppressor function, while also conferring dominant-negative and gain-of-function oncogenic properties including promotion of invasion, metastasis, and MDM2 antagonism (COSMIC ID COSM10648). In OSCC, TP53 mutations are present in 50–70% of cases and are associated with aggressive tumour behaviour, locoregional recurrence, and reduced sensitivity to platinum-based chemotherapy. The VAF of 0.60% in a surveillance swab warrants clinical correlation to exclude early local recurrence. Imaging and clinical re-evaluation are recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000x raw input depth (≥400x consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 2,000x raw sequencing depth, approximately 15–20% of targets may fall below 100x consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Dickens, Charles J.

Accession:

NGS-2026-052103

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR			Cov ✓
L858R	c.2573T>G	Not detected	
T790M	c.2369C>T	Not detected	
Exon 19 del	c.2235_2249del	Not detected	
KRAS			Cov ✓
G12D	c.35G>T	Not detected	
G12V	c.35G>A	Not detected	
G12C	c.34G>T	Not detected	
G13D	c.38G>A	Not detected	
PIK3CA			Cov ✓
E542K	c.1624G>A	Not detected	
E545K	c.1633G>A	Not detected	
H1047R	c.3140A>G	Not detected	
TERT			Cov ✓
C228T	c.-124C>T	Not detected	
C250T	c.-146C>T	Not detected	

HRAS			Cov ✓
G12V	c.35G>T	Not detected	
G12D	c.35G>A	Not detected	
Q61L	c.182A>T	Not detected	
Q61R	c.182A>G	Not detected	
NOTCH1			Cov ✓
P2415L	c.7244C>T	Not detected	
R1699W	c.5095C>T	Not detected	
TP53			Cov ✓
R175H	c.524G>A	● DETECTED	
R248Q	c.743G>A	Not detected	
R248W	c.742C>T	Not detected	
R273H	c.818G>A	Not detected	
R282W	c.844C>T	Not detected	

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)

251 targets, 77,417 bp total. Mean consensus depth: 376x. Targets below 100x consensus: 72 / 251 (29%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	178x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	362x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	79x	⚠ Below 100x threshold
DPYD	chr1:97,573,763-97,573,963	201	592x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	85x	⚠ Below 100x threshold
PTEN	chr10:87,894,005-87,894,129	125	52x	⚠ Below 100x threshold
PTEN	chr10:87,925,493-87,925,577	85	269x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	260x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	300x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	1,179x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	132x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	53x	⚠ Below 100x threshold
PTEN	chr10:87,965,267-87,965,489	223	382x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	199x	✓ Covered
HRAS	chr11:533,433-533,632	200	692x	✓ Covered
HRAS	chr11:533,746-533,964	219	753x	✓ Covered
HRAS	chr11:534,192-534,342	151	1,313x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	72x	⚠ Below 100x threshold
KRAS	chr12:25,225,594-25,225,793	200	1,011x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	191x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	583x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	266x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	77x	⚠ Below 100x threshold
KMT2D	chr12:49,022,260-49,022,373	114	1,576x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	119x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	53x	⚠ Below 100x threshold
KMT2D	chr12:49,024,790-49,024,966	177	61x	⚠ Below 100x threshold
KMT2D	chr12:49,026,162-49,027,342	1,181	87x	⚠ Below 100x threshold
KMT2D	chr12:49,027,783-49,027,950	168	279x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	557x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	130x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	783x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	251x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	98x	⚠ Below 100x threshold
KMT2D	chr12:49,030,581-49,030,788	208	571x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	1,967x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	104x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	85x	⚠ Below 100x threshold
KMT2D	chr12:49,034,390-49,034,496	107	328x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	152x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	843x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	701x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	893x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	134x	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	411×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	117×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	22×	⚠ Below 100× threshold
KMT2D	chr12:49,042,069–49,042,350	282	509×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	84×	⚠ Below 100× threshold
KMT2D	chr12:49,042,721–49,042,898	178	86×	⚠ Below 100× threshold
KMT2D	chr12:49,043,056–49,043,206	151	107×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	192×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	77×	⚠ Below 100× threshold
KMT2D	chr12:49,043,848–49,044,018	171	161×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	262×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	144×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	326×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	125×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	13×	⚠ Below 100× threshold
KMT2D	chr12:49,046,240–49,046,444	205	275×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	96×	⚠ Below 100× threshold
KMT2D	chr12:49,047,945–49,048,089	145	266×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	203×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	296×	✓ Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	784×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	69×	⚠ Below 100× threshold
KMT2D	chr12:49,052,544–49,052,729	186	1,290×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	1,697×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	87×	⚠ Below 100× threshold
KMT2D	chr12:49,053,456–49,053,661	206	82×	⚠ Below 100× threshold
KMT2D	chr12:49,053,958–49,054,160	203	148×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	705×	✓ Covered
KMT2D	chr12:49,054,508–49,054,771	264	225×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	256×	✓ Covered
KMT2D	chr12:49,055,256–49,055,344	89	303×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	3,363×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	214×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	329×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	88×	⚠ Below 100× threshold
AJUBA	chr14:22,976,436–22,976,538	103	184×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	544×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	23×	⚠ Below 100× threshold
AJUBA	chr14:22,981,241–22,982,286	1,046	348×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	259×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	48×	⚠ Below 100× threshold
TP53	chr17:7,673,515–7,673,628	114	178×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	238×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	121×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	45×	⚠ Below 100× threshold
TP53	chr17:7,675,033–7,675,256	224	255×	● VARIANT DETECTED
TP53	chr17:7,675,974–7,676,292	319	142×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	809×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	60×	⚠ Below 100× threshold

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	172x	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	194x	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	105x	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	663x	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	450x	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	483x	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	507x	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	455x	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	47x	⚠ Below 100x threshold
CASP8	chr2:201,272,878–201,272,962	85	526x	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	728x	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	463x	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	1,167x	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	845x	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	156x	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	414x	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	92x	⚠ Below 100x threshold
CUL3	chr2:224,495,812–224,495,986	175	11x	⚠ Below 100x threshold
CUL3	chr2:224,497,733–224,497,869	137	94x	⚠ Below 100x threshold
CUL3	chr2:224,500,343–224,500,507	165	2,004x	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	35x	⚠ Below 100x threshold
CUL3	chr2:224,503,632–224,503,842	211	52x	⚠ Below 100x threshold
CUL3	chr2:224,505,936–224,506,152	217	272x	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	315x	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	100x	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	83x	⚠ Below 100x threshold
CUL3	chr2:224,514,592–224,514,792	201	44x	⚠ Below 100x threshold
CUL3	chr2:224,535,508–224,535,661	154	264x	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	541x	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	161x	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	832x	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	27x	⚠ Below 100x threshold
EP300	chr22:41,125,844–41,126,060	217	89x	⚠ Below 100x threshold
EP300	chr22:41,127,467–41,127,768	302	88x	⚠ Below 100x threshold
EP300	chr22:41,129,870–41,130,023	154	93x	⚠ Below 100x threshold
EP300	chr22:41,131,368–41,131,653	286	133x	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	84x	⚠ Below 100x threshold
EP300	chr22:41,137,633–41,137,810	178	1,097x	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	114x	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	80x	⚠ Below 100x threshold
EP300	chr22:41,146,719–41,146,836	118	1,081x	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	37x	⚠ Below 100x threshold
EP300	chr22:41,149,018–41,149,195	178	19x	⚠ Below 100x threshold
EP300	chr22:41,149,741–41,150,218	478	53x	⚠ Below 100x threshold
EP300	chr22:41,151,813–41,152,032	220	52x	⚠ Below 100x threshold
EP300	chr22:41,152,186–41,152,370	185	175x	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	292×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	89×	⚠ Below 100× threshold
EP300	chr22:41,158,392–41,158,520	129	695×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	1,352×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	103×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	25×	⚠ Below 100× threshold
EP300	chr22:41,166,579–41,166,686	108	381×	✓ Covered
EP300	chr22:41,168,429–41,168,619	191	262×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	36×	⚠ Below 100× threshold
EP300	chr22:41,169,483–41,169,636	154	232×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	73×	⚠ Below 100× threshold
EP300	chr22:41,172,479–41,172,683	205	365×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	1,349×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	671×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	2,019×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	60×	⚠ Below 100× threshold
PIK3CA	chr3:179,234,074–179,234,381	308	612×	✓ Covered
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	86×	⚠ Below 100× threshold
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	67×	⚠ Below 100× threshold
FAT1	chr4:186,595,669–186,595,846	178	808×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	74×	⚠ Below 100× threshold
FAT1	chr4:186,597,662–186,597,812	151	250×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	113×	✓ Covered
FAT1	chr4:186,599,878–186,600,380	503	502×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	109×	✓ Covered
FAT1	chr4:186,602,883–186,603,054	172	286×	✓ Covered
FAT1	chr4:186,603,156–186,603,997	842	247×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	385×	✓ Covered
FAT1	chr4:186,606,050–186,606,233	184	293×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	25×	⚠ Below 100× threshold
FAT1	chr4:186,609,781–186,610,035	255	152×	✓ Covered
FAT1	chr4:186,611,366–186,611,795	430	126×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	25×	⚠ Below 100× threshold
FAT1	chr4:186,614,171–186,614,364	194	133×	✓ Covered
FAT1	chr4:186,616,985–186,617,221	237	296×	✓ Covered
FAT1	chr4:186,617,688–186,621,795	4,108	218×	✓ Covered
FAT1	chr4:186,628,134–186,628,384	251	129×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	257×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	109×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	393×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	75×	⚠ Below 100× threshold
FAT1	chr4:186,639,702–186,639,803	102	40×	⚠ Below 100× threshold
FAT1	chr4:186,663,279–186,663,633	355	291×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	304×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	5,245×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	430×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	1,330×	✓ Covered
PIK3R1	chr5:68,295,400-68,295,508	109	198×	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	1,142×	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	89×	⚠ Below 100× threshold
NSD1	chr5:177,204,100-177,204,312	213	792×	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	423×	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	431×	✓ Covered
NSD1	chr5:177,238,217-177,238,527	311	21×	⚠ Below 100× threshold
NSD1	chr5:177,239,736-177,239,885	150	1,008×	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	182×	✓ Covered
NSD1	chr5:177,246,658-177,246,816	159	111×	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	612×	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	92×	⚠ Below 100× threshold
NSD1	chr5:177,256,931-177,257,171	241	388×	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	64×	⚠ Below 100× threshold
NSD1	chr5:177,267,542-177,267,738	197	348×	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	198×	✓ Covered
NSD1	chr5:177,273,652-177,273,804	153	201×	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	239×	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	83×	⚠ Below 100× threshold
NSD1	chr5:177,283,767-177,283,948	182	52×	⚠ Below 100× threshold
NSD1	chr5:177,288,799-177,288,945	147	1,193×	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	507×	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	168×	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	615×	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	59×	⚠ Below 100× threshold
EGFR	chr7:55,181,273-55,181,498	226	97×	⚠ Below 100× threshold
EGFR	chr7:55,191,699-55,191,894	196	59×	⚠ Below 100× threshold
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	1,351×	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	139×	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	1,178×	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	118×	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	218×	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	59×	⚠ Below 100× threshold
NOTCH1	chr9:136,499,092-136,499,279	188	512×	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	357×	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	133×	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	232×	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	152×	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	625×	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	425×	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	31×	⚠ Below 100× threshold
NOTCH1	chr9:136,506,507-136,506,659	153	574×	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	618×	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	42×	⚠ Below 100× threshold
NOTCH1	chr9:136,507,935-136,508,159	225	364×	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	345×	✓ Covered

FULL PANEL COVERAGE REPORT — Dickens, Charles J. (NGS-2026-052103)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	194x	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	44x	⚠ Below 100x threshold
NOTCH1	chr9:136,510,633-136,510,825	193	82x	⚠ Below 100x threshold
NOTCH1	chr9:136,511,132-136,511,291	160	145x	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	238x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	71x	⚠ Below 100x threshold
NOTCH1	chr9:136,514,490-136,514,722	233	27x	⚠ Below 100x threshold
NOTCH1	chr9:136,515,270-136,515,420	151	939x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	193x	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	353x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	377x	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	152x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	245x	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	280x	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	142x	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	1,029x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	34x	⚠ Below 100x threshold
NOTCH1	chr9:136,544,004-136,544,122	119	180x	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	437x	✓ Covered

PATIENT & SPECIMEN INFORMATION			
Patient name	Kafka, Franz J.	Accession	NGS-2026-052105
Date of birth	1980-02-17 (Age 46)	Specimen	Oral swab, anterior tongue
MRN	50789065	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — anterior tongue (surveillance post-resection)		2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at $\geq 0.2\%$ VAF with ≥ 2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤ 1 edit distance), and molecular consensus sequences are called requiring ≥ 2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants $\geq 0.2\%$ consensus VAF with ≥ 2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF $\geq 0.2\%$ with validated sensitivity of 98.4% at $\geq 20,000\times$ raw input depth ($\geq 400\times$ consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At $2,000\times$ raw sequencing depth, approximately 15–20% of targets may fall below $100\times$ consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient: Kafka, Franz J.

Accession: NGS-2026-052105

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	Not detected
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)

251 targets, 77,417 bp total. Mean consensus depth: 426x. Targets below 100x consensus: 60 / 251 (24%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	62x	⚠ Below 100x threshold
DPYD	chr1:97,449,958-97,450,158	201	167x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	196x	✓ Covered
DPYD	chr1:97,573,763-97,573,963	201	1,256x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	191x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	823x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	580x	✓ Covered
PTEN	chr10:87,931,026-87,931,109	84	545x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	389x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	42x	⚠ Below 100x threshold
PTEN	chr10:87,957,833-87,958,039	207	224x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	266x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	459x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	240x	✓ Covered
HRAS	chr11:533,433-533,632	200	61x	⚠ Below 100x threshold
HRAS	chr11:533,746-533,964	219	82x	⚠ Below 100x threshold
HRAS	chr11:534,192-534,342	151	54x	⚠ Below 100x threshold
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	150x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	222x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	1,628x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	43x	⚠ Below 100x threshold
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	94x	⚠ Below 100x threshold
KMT2D	chr12:49,022,023-49,022,171	149	115x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	644x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	1,703x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	300x	✓ Covered
KMT2D	chr12:49,024,790-49,024,966	177	199x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	548x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	61x	⚠ Below 100x threshold
KMT2D	chr12:49,027,989-49,028,161	173	72x	⚠ Below 100x threshold
KMT2D	chr12:49,028,808-49,028,978	171	158x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	1,547x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	68x	⚠ Below 100x threshold
KMT2D	chr12:49,030,260-49,030,459	200	96x	⚠ Below 100x threshold
KMT2D	chr12:49,030,581-49,030,788	208	24x	⚠ Below 100x threshold
KMT2D	chr12:49,030,873-49,031,053	181	277x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	151x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	112x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	91x	⚠ Below 100x threshold
KMT2D	chr12:49,034,562-49,034,686	125	329x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	2,066x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	112x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	430x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	46x	⚠ Below 100x threshold

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KMT2D	chr12:49,039,704–49,041,555	1,852	2,470×	✓ Covered
KMT2D	chr12:49,041,635–49,041,725	91	1,752×	✓ Covered
KMT2D	chr12:49,041,897–49,042,010	114	498×	✓ Covered
KMT2D	chr12:49,042,069–49,042,350	282	1,204×	✓ Covered
KMT2D	chr12:49,042,541–49,042,665	125	250×	✓ Covered
KMT2D	chr12:49,042,721–49,042,898	178	267×	✓ Covered
KMT2D	chr12:49,043,056–49,043,206	151	389×	✓ Covered
KMT2D	chr12:49,043,343–49,043,448	106	2,091×	✓ Covered
KMT2D	chr12:49,043,615–49,043,802	188	180×	✓ Covered
KMT2D	chr12:49,043,848–49,044,018	171	257×	✓ Covered
KMT2D	chr12:49,044,180–49,044,324	145	924×	✓ Covered
KMT2D	chr12:49,044,383–49,044,542	160	439×	✓ Covered
KMT2D	chr12:49,044,724–49,044,985	262	1,024×	✓ Covered
KMT2D	chr12:49,045,900–49,045,987	88	994×	✓ Covered
KMT2D	chr12:49,046,045–49,046,194	150	69×	⚠ Below 100× threshold
KMT2D	chr12:49,046,240–49,046,444	205	115×	✓ Covered
KMT2D	chr12:49,046,589–49,046,810	222	238×	✓ Covered
KMT2D	chr12:49,047,945–49,048,089	145	190×	✓ Covered
KMT2D	chr12:49,048,639–49,048,789	151	324×	✓ Covered
KMT2D	chr12:49,049,085–49,049,238	154	56×	⚠ Below 100× threshold
KMT2D	chr12:49,049,662–49,050,810	1,149	431×	✓ Covered
KMT2D	chr12:49,050,866–49,052,444	1,579	365×	✓ Covered
KMT2D	chr12:49,052,544–49,052,729	186	268×	✓ Covered
KMT2D	chr12:49,052,895–49,053,092	198	469×	✓ Covered
KMT2D	chr12:49,053,187–49,053,341	155	87×	⚠ Below 100× threshold
KMT2D	chr12:49,053,456–49,053,661	206	189×	✓ Covered
KMT2D	chr12:49,053,958–49,054,160	203	112×	✓ Covered
KMT2D	chr12:49,054,287–49,054,436	150	28×	⚠ Below 100× threshold
KMT2D	chr12:49,054,508–49,054,771	264	1,250×	✓ Covered
KMT2D	chr12:49,054,880–49,055,046	167	87×	⚠ Below 100× threshold
KMT2D	chr12:49,055,256–49,055,344	89	262×	✓ Covered
AJUBA				
AJUBA	chr14:22,973,426–22,973,588	163	212×	✓ Covered
AJUBA	chr14:22,974,027–22,974,135	109	1,275×	✓ Covered
AJUBA	chr14:22,974,819–22,974,910	92	108×	✓ Covered
AJUBA	chr14:22,974,954–22,975,124	171	720×	✓ Covered
AJUBA	chr14:22,976,436–22,976,538	103	350×	✓ Covered
AJUBA	chr14:22,976,625–22,976,732	108	102×	✓ Covered
AJUBA	chr14:22,978,324–22,978,465	142	20×	⚠ Below 100× threshold
AJUBA	chr14:22,981,241–22,982,286	1,046	642×	✓ Covered
TP53				
TP53	chr17:7,669,592–7,669,710	119	993×	✓ Covered
TP53	chr17:7,670,589–7,670,735	147	121×	✓ Covered
TP53	chr17:7,673,515–7,673,628	114	211×	✓ Covered
TP53	chr17:7,673,681–7,673,857	177	491×	✓ Covered
TP53	chr17:7,674,161–7,674,310	150	399×	✓ Covered
TP53	chr17:7,674,839–7,674,991	153	257×	✓ Covered
TP53	chr17:7,675,033–7,675,256	224	240×	✓ Covered
TP53	chr17:7,675,974–7,676,292	319	251×	✓ Covered
TP53	chr17:7,676,362–7,676,423	62	348×	✓ Covered
TP53	chr17:7,676,501–7,676,614	114	132×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	69×	⚠ Below 100× threshold
KEAP1	chr19:10,489,172–10,489,388	217	123×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	179×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	361×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	220×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	224×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	166×	✓ Covered
CASP8	chr2:201,271,496–201,271,641	146	114×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	345×	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	40×	⚠ Below 100× threshold
CASP8	chr2:201,274,869–201,274,973	105	385×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	604×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	122×	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	273×	✓ Covered
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	337×	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	1,795×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	132×	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	163×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	206×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	377×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	196×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	3,481×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	84×	⚠ Below 100× threshold
CUL3	chr2:224,506,838–224,507,023	186	532×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	127×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	102×	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	345×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	154×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	181×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	78×	⚠ Below 100× threshold
EP300				
EP300	chr22:41,092,985–41,093,118	134	407×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	193×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	23×	⚠ Below 100× threshold
EP300	chr22:41,127,467–41,127,768	302	846×	✓ Covered
EP300	chr22:41,129,870–41,130,023	154	116×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	761×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	178×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	36×	⚠ Below 100× threshold
EP300	chr22:41,140,120–41,140,277	158	174×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	41×	⚠ Below 100× threshold
EP300	chr22:41,146,719–41,146,836	118	1,633×	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	79×	⚠ Below 100× threshold
EP300	chr22:41,149,018–41,149,195	178	1,397×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	101×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	118×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	331×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975–41,155,133	159	289×	✓ Covered
EP300	chr22:41,157,149–41,157,428	280	346×	✓ Covered
EP300	chr22:41,158,392–41,158,520	129	238×	✓ Covered
EP300	chr22:41,160,622–41,160,742	121	381×	✓ Covered
EP300	chr22:41,162,703–41,162,799	97	1,543×	✓ Covered
EP300	chr22:41,164,033–41,164,150	118	925×	✓ Covered
EP300	chr22:41,166,579–41,166,686	108	6×	⚠ Below 100× threshold
EP300	chr22:41,168,429–41,168,619	191	914×	✓ Covered
EP300	chr22:41,168,701–41,168,887	187	119×	✓ Covered
EP300	chr22:41,169,483–41,169,636	154	1,073×	✓ Covered
EP300	chr22:41,170,386–41,170,591	206	158×	✓ Covered
EP300	chr22:41,172,479–41,172,683	205	113×	✓ Covered
EP300	chr22:41,173,603–41,173,804	202	108×	✓ Covered
EP300	chr22:41,176,227–41,176,548	322	348×	✓ Covered
EP300	chr22:41,176,753–41,178,973	2,221	43×	⚠ Below 100× threshold
PIK3CA				
PIK3CA	chr3:179,218,190–179,218,354	165	75×	⚠ Below 100× threshold
PIK3CA	chr3:179,234,074–179,234,381	308	59×	⚠ Below 100× threshold
FBXW7				
FBXW7	chr4:152,325,986–152,326,251	266	133×	✓ Covered
FAT1				
FAT1	chr4:186,588,575–186,589,240	666	60×	⚠ Below 100× threshold
FAT1	chr4:186,595,669–186,595,846	178	226×	✓ Covered
FAT1	chr4:186,596,520–186,597,191	672	92×	⚠ Below 100× threshold
FAT1	chr4:186,597,662–186,597,812	151	1,980×	✓ Covered
FAT1	chr4:186,597,952–186,598,145	194	28×	⚠ Below 100× threshold
FAT1	chr4:186,599,878–186,600,380	503	963×	✓ Covered
FAT1	chr4:186,601,249–186,601,446	198	87×	⚠ Below 100× threshold
FAT1	chr4:186,602,883–186,603,054	172	54×	⚠ Below 100× threshold
FAT1	chr4:186,603,156–186,603,997	842	409×	✓ Covered
FAT1	chr4:186,604,357–186,604,594	238	72×	⚠ Below 100× threshold
FAT1	chr4:186,606,050–186,606,233	184	946×	✓ Covered
FAT1	chr4:186,609,163–186,609,340	178	400×	✓ Covered
FAT1	chr4:186,609,781–186,610,035	255	85×	⚠ Below 100× threshold
FAT1	chr4:186,611,366–186,611,795	430	102×	✓ Covered
FAT1	chr4:186,613,089–186,613,362	274	112×	✓ Covered
FAT1	chr4:186,614,171–186,614,364	194	46×	⚠ Below 100× threshold
FAT1	chr4:186,616,985–186,617,221	237	87×	⚠ Below 100× threshold
FAT1	chr4:186,617,688–186,621,795	4,108	61×	⚠ Below 100× threshold
FAT1	chr4:186,628,134–186,628,384	251	736×	✓ Covered
FAT1	chr4:186,628,468–186,628,783	316	367×	✓ Covered
FAT1	chr4:186,633,664–186,633,843	180	537×	✓ Covered
FAT1	chr4:186,636,005–186,636,255	251	2,151×	✓ Covered
FAT1	chr4:186,636,565–186,636,934	370	121×	✓ Covered
FAT1	chr4:186,639,702–186,639,803	102	42×	⚠ Below 100× threshold
FAT1	chr4:186,663,279–186,663,633	355	2,038×	✓ Covered
FAT1	chr4:186,706,543–186,709,847	3,305	135×	✓ Covered
TERT				
TERT	chr5:1,295,050–1,295,250	201	267×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516–68,294,698	183	142×	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	94x	⚠ Below 100x threshold
PIK3R1	chr5:68,295,400-68,295,508	109	281x	✓ Covered
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	114x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	887x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	855x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	30x	⚠ Below 100x threshold
NSD1	chr5:177,235,801-177,235,965	165	77x	⚠ Below 100x threshold
NSD1	chr5:177,238,217-177,238,527	311	477x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	169x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	16x	⚠ Below 100x threshold
NSD1	chr5:177,246,658-177,246,816	159	94x	⚠ Below 100x threshold
NSD1	chr5:177,248,161-177,248,344	184	974x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	134x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	288x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	957x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	56x	⚠ Below 100x threshold
NSD1	chr5:177,269,582-177,269,827	246	37x	⚠ Below 100x threshold
NSD1	chr5:177,273,652-177,273,804	153	22x	⚠ Below 100x threshold
NSD1	chr5:177,280,545-177,280,854	310	154x	✓ Covered
NSD1	chr5:177,282,445-177,282,601	157	236x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	134x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	239x	✓ Covered
NSD1	chr5:177,291,934-177,292,178	245	896x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	3,548x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	40x	⚠ Below 100x threshold
EGFR	chr7:55,174,702-55,174,840	139	97x	⚠ Below 100x threshold
EGFR	chr7:55,181,273-55,181,498	226	495x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	65x	⚠ Below 100x threshold
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	124x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	409x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	245x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,847	190	409x	✓ Covered
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	364x	✓ Covered
NOTCH1	chr9:136,498,879-136,499,016	138	20x	⚠ Below 100x threshold
NOTCH1	chr9:136,499,092-136,499,279	188	562x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	513x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	1,886x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	455x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	294x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	115x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	129x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	882x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	75x	⚠ Below 100x threshold
NOTCH1	chr9:136,506,696-136,506,993	298	165x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	110x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	425x	✓ Covered
NOTCH1	chr9:136,508,212-136,508,405	194	111x	✓ Covered

FULL PANEL COVERAGE REPORT — Kafka, Franz J. (NGS-2026-052105)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	2,182×	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	592×	✓ Covered
NOTCH1	chr9:136,510,633-136,510,825	193	284×	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	77×	⚠ Below 100× threshold
NOTCH1	chr9:136,513,001-136,513,154	154	92×	⚠ Below 100× threshold
NOTCH1	chr9:136,513,372-136,513,557	186	326×	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	135×	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	98×	⚠ Below 100× threshold
NOTCH1	chr9:136,515,463-136,515,736	274	344×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	1,630×	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	927×	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	122×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	590×	✓ Covered
NOTCH1	chr9:136,518,571-136,518,844	274	434×	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	84×	⚠ Below 100× threshold
NOTCH1	chr9:136,522,830-136,523,208	379	190×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	180×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	1,621×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	158×	✓ Covered

PATIENT & SPECIMEN INFORMATION			
Patient name	Morrison, Toni E.	Accession	NGS-2026-052106
Date of birth	1955-10-09 (Age 70)	Specimen	Oral swab, left retromolar trigone
MRN	60890176	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — left retromolar trigone	Reported	2026-05-20

RESULT: NO REPORTABLE VARIANTS DETECTED

INTERPRETATION

No pathogenic or likely pathogenic somatic variants were detected in the 21 genes covered by this assay at $\geq 0.2\%$ VAF with ≥ 2 consensus supporting reads. A negative result does not exclude malignancy. Variants below the reporting threshold, variants outside the 251 assay target regions, structural variants, and copy number alterations are not assessed by this assay. Clinical correlation is required; repeat testing or biopsy should be considered if clinical suspicion persists.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53

Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2x150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤ 1 edit distance), and molecular consensus sequences are called requiring ≥ 2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants $\geq 0.2\%$ consensus VAF with ≥ 2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF $\geq 0.2\%$ with validated sensitivity of 98.4% at $\geq 20,000\times$ raw input depth ($\geq 400\times$ consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At $2,000\times$ raw sequencing depth, approximately 15–20% of targets may fall below $100\times$ consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Morrison, Toni E.

Accession:

NGS-2026-052106

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFRCov ✓			HRASCov ✓		
L858R	c.2573T>G	Not detected	G12V	c.35G>T	Not detected
T790M	c.2369C>T	Not detected	G12D	c.35G>A	Not detected
Exon 19 del	c.2235_2249del	Not detected	Q61L	c.182A>T	Not detected
			Q61R	c.182A>G	Not detected
KRASCov ✓			NOTCH1Cov ✓		
G12D	c.35G>T	Not detected	P2415L	c.7244C>T	Not detected
G12V	c.35G>A	Not detected	R1699W	c.5095C>T	Not detected
G12C	c.34G>T	Not detected			
G13D	c.38G>A	Not detected	TP53Cov ✓		
PIK3CACov ✓			R175H	c.524G>A	Not detected
E542K	c.1624G>A	Not detected	R248Q	c.743G>A	Not detected
E545K	c.1633G>A	Not detected	R248W	c.742C>T	Not detected
H1047R	c.3140A>G	Not detected	R273H	c.818G>A	Not detected
TERTCov ✓			R282W	c.844C>T	Not detected
C228T	c.-124C>T	Not detected			
C250T	c.-146C>T	Not detected			

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)

251 targets, 77,417 bp total. Mean consensus depth: 353x. Targets below 100x consensus: 70 / 251 (28%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	482x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	168x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	67x	⚠ Below 100x threshold
DPYD	chr1:97,573,763-97,573,963	201	60x	⚠ Below 100x threshold
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	33x	⚠ Below 100x threshold
PTEN	chr10:87,894,005-87,894,129	125	181x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	72x	⚠ Below 100x threshold
PTEN	chr10:87,931,026-87,931,109	84	649x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	53x	⚠ Below 100x threshold
PTEN	chr10:87,952,098-87,952,279	182	264x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	317x	✓ Covered
PTEN	chr10:87,960,874-87,961,138	265	257x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	282x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	1,395x	✓ Covered
HRAS	chr11:533,433-533,632	200	91x	⚠ Below 100x threshold
HRAS	chr11:533,746-533,964	219	326x	✓ Covered
HRAS	chr11:534,192-534,342	151	376x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	54x	⚠ Below 100x threshold
KRAS	chr12:25,225,594-25,225,793	200	359x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	392x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	459x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	186x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	403x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	321x	✓ Covered
KMT2D	chr12:49,022,570-49,022,895	326	668x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	83x	⚠ Below 100x threshold
KMT2D	chr12:49,024,790-49,024,966	177	743x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	53x	⚠ Below 100x threshold
KMT2D	chr12:49,027,783-49,027,950	168	171x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	297x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	791x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	131x	✓ Covered
KMT2D	chr12:49,029,381-49,029,496	116	399x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	192x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	2,651x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	252x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	251x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	75x	⚠ Below 100x threshold
KMT2D	chr12:49,034,390-49,034,496	107	30x	⚠ Below 100x threshold
KMT2D	chr12:49,034,562-49,034,686	125	115x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	75x	⚠ Below 100x threshold
KMT2D	chr12:49,037,105-49,039,009	1,905	275x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	220x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	783x	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)					
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status	
KMT2D	chr12:49,039,704–49,041,555	1,852	295×	✓	Covered
KMT2D	chr12:49,041,635–49,041,725	91	112×	✓	Covered
KMT2D	chr12:49,041,897–49,042,010	114	578×	✓	Covered
KMT2D	chr12:49,042,069–49,042,350	282	140×	✓	Covered
KMT2D	chr12:49,042,541–49,042,665	125	133×	✓	Covered
KMT2D	chr12:49,042,721–49,042,898	178	135×	✓	Covered
KMT2D	chr12:49,043,056–49,043,206	151	668×	✓	Covered
KMT2D	chr12:49,043,343–49,043,448	106	28×	⚠	Below 100× threshold
KMT2D	chr12:49,043,615–49,043,802	188	147×	✓	Covered
KMT2D	chr12:49,043,848–49,044,018	171	192×	✓	Covered
KMT2D	chr12:49,044,180–49,044,324	145	208×	✓	Covered
KMT2D	chr12:49,044,383–49,044,542	160	342×	✓	Covered
KMT2D	chr12:49,044,724–49,044,985	262	72×	⚠	Below 100× threshold
KMT2D	chr12:49,045,900–49,045,987	88	176×	✓	Covered
KMT2D	chr12:49,046,045–49,046,194	150	284×	✓	Covered
KMT2D	chr12:49,046,240–49,046,444	205	326×	✓	Covered
KMT2D	chr12:49,046,589–49,046,810	222	1,765×	✓	Covered
KMT2D	chr12:49,047,945–49,048,089	145	215×	✓	Covered
KMT2D	chr12:49,048,639–49,048,789	151	359×	✓	Covered
KMT2D	chr12:49,049,085–49,049,238	154	258×	✓	Covered
KMT2D	chr12:49,049,662–49,050,810	1,149	25×	⚠	Below 100× threshold
KMT2D	chr12:49,050,866–49,052,444	1,579	263×	✓	Covered
KMT2D	chr12:49,052,544–49,052,729	186	170×	✓	Covered
KMT2D	chr12:49,052,895–49,053,092	198	218×	✓	Covered
KMT2D	chr12:49,053,187–49,053,341	155	975×	✓	Covered
KMT2D	chr12:49,053,456–49,053,661	206	171×	✓	Covered
KMT2D	chr12:49,053,958–49,054,160	203	204×	✓	Covered
KMT2D	chr12:49,054,287–49,054,436	150	10×	⚠	Below 100× threshold
KMT2D	chr12:49,054,508–49,054,771	264	219×	✓	Covered
KMT2D	chr12:49,054,880–49,055,046	167	5,080×	✓	Covered
KMT2D	chr12:49,055,256–49,055,344	89	193×	✓	Covered
AJUBA					
AJUBA	chr14:22,973,426–22,973,588	163	2,225×	✓	Covered
AJUBA	chr14:22,974,027–22,974,135	109	132×	✓	Covered
AJUBA	chr14:22,974,819–22,974,910	92	141×	✓	Covered
AJUBA	chr14:22,974,954–22,975,124	171	470×	✓	Covered
AJUBA	chr14:22,976,436–22,976,538	103	258×	✓	Covered
AJUBA	chr14:22,976,625–22,976,732	108	159×	✓	Covered
AJUBA	chr14:22,978,324–22,978,465	142	86×	⚠	Below 100× threshold
AJUBA	chr14:22,981,241–22,982,286	1,046	82×	⚠	Below 100× threshold
TP53					
TP53	chr17:7,669,592–7,669,710	119	1,339×	✓	Covered
TP53	chr17:7,670,589–7,670,735	147	96×	⚠	Below 100× threshold
TP53	chr17:7,673,515–7,673,628	114	153×	✓	Covered
TP53	chr17:7,673,681–7,673,857	177	126×	✓	Covered
TP53	chr17:7,674,161–7,674,310	150	106×	✓	Covered
TP53	chr17:7,674,839–7,674,991	153	310×	✓	Covered
TP53	chr17:7,675,033–7,675,256	224	986×	✓	Covered
TP53	chr17:7,675,974–7,676,292	319	97×	⚠	Below 100× threshold
TP53	chr17:7,676,362–7,676,423	62	109×	✓	Covered
TP53	chr17:7,676,501–7,676,614	114	126×	✓	Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	210×	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	199×	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	329×	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	106×	✓ Covered
KEAP1	chr19:10,499,375–10,500,053	679	998×	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	124×	✓ Covered
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	3×	⚠ Below 100× threshold
CASP8	chr2:201,271,496–201,271,641	146	141×	✓ Covered
CASP8	chr2:201,272,618–201,272,796	179	88×	⚠ Below 100× threshold
CASP8	chr2:201,272,878–201,272,962	85	55×	⚠ Below 100× threshold
CASP8	chr2:201,274,869–201,274,973	105	837×	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	294×	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	74×	⚠ Below 100× threshold
CASP8	chr2:201,286,439–201,286,611	173	74×	⚠ Below 100× threshold
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	56×	⚠ Below 100× threshold
CUL3	chr2:224,478,180–224,478,365	186	257×	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	50×	⚠ Below 100× threshold
CUL3	chr2:224,495,812–224,495,986	175	109×	✓ Covered
CUL3	chr2:224,497,733–224,497,869	137	180×	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	624×	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	271×	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	677×	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	149×	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	252×	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	884×	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	35×	⚠ Below 100× threshold
CUL3	chr2:224,514,592–224,514,792	201	316×	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	154×	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	392×	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	173×	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	751×	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	298×	✓ Covered
EP300	chr22:41,125,844–41,126,060	217	1,533×	✓ Covered
EP300	chr22:41,127,467–41,127,768	302	70×	⚠ Below 100× threshold
EP300	chr22:41,129,870–41,130,023	154	700×	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	1,229×	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	252×	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	269×	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	304×	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	949×	✓ Covered
EP300	chr22:41,146,719–41,146,836	118	55×	⚠ Below 100× threshold
EP300	chr22:41,147,817–41,147,966	150	176×	✓ Covered
EP300	chr22:41,149,018–41,149,195	178	134×	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	185×	✓ Covered
EP300	chr22:41,151,813–41,152,032	220	281×	✓ Covered
EP300	chr22:41,152,186–41,152,370	185	717×	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	16× ⚠ Below 100× threshold	
EP300	chr22:41,157,149-41,157,428	280	87× ⚠ Below 100× threshold	
EP300	chr22:41,158,392-41,158,520	129	324× ✓ Covered	
EP300	chr22:41,160,622-41,160,742	121	304× ✓ Covered	
EP300	chr22:41,162,703-41,162,799	97	494× ✓ Covered	
EP300	chr22:41,164,033-41,164,150	118	2,608× ✓ Covered	
EP300	chr22:41,166,579-41,166,686	108	85× ⚠ Below 100× threshold	
EP300	chr22:41,168,429-41,168,619	191	410× ✓ Covered	
EP300	chr22:41,168,701-41,168,887	187	41× ⚠ Below 100× threshold	
EP300	chr22:41,169,483-41,169,636	154	853× ✓ Covered	
EP300	chr22:41,170,386-41,170,591	206	394× ✓ Covered	
EP300	chr22:41,172,479-41,172,683	205	78× ⚠ Below 100× threshold	
EP300	chr22:41,173,603-41,173,804	202	136× ✓ Covered	
EP300	chr22:41,176,227-41,176,548	322	36× ⚠ Below 100× threshold	
EP300	chr22:41,176,753-41,178,973	2,221	328× ✓ Covered	
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	39× ⚠ Below 100× threshold	
PIK3CA	chr3:179,234,074-179,234,381	308	83× ⚠ Below 100× threshold	
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	351× ✓ Covered	
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	230× ✓ Covered	
FAT1	chr4:186,595,669-186,595,846	178	228× ✓ Covered	
FAT1	chr4:186,596,520-186,597,191	672	813× ✓ Covered	
FAT1	chr4:186,597,662-186,597,812	151	197× ✓ Covered	
FAT1	chr4:186,597,952-186,598,145	194	342× ✓ Covered	
FAT1	chr4:186,599,878-186,600,380	503	42× ⚠ Below 100× threshold	
FAT1	chr4:186,601,249-186,601,446	198	451× ✓ Covered	
FAT1	chr4:186,602,883-186,603,054	172	103× ✓ Covered	
FAT1	chr4:186,603,156-186,603,997	842	63× ⚠ Below 100× threshold	
FAT1	chr4:186,604,357-186,604,594	238	194× ✓ Covered	
FAT1	chr4:186,606,050-186,606,233	184	85× ⚠ Below 100× threshold	
FAT1	chr4:186,609,163-186,609,340	178	24× ⚠ Below 100× threshold	
FAT1	chr4:186,609,781-186,610,035	255	181× ✓ Covered	
FAT1	chr4:186,611,366-186,611,795	430	1,619× ✓ Covered	
FAT1	chr4:186,613,089-186,613,362	274	190× ✓ Covered	
FAT1	chr4:186,614,171-186,614,364	194	1,340× ✓ Covered	
FAT1	chr4:186,616,985-186,617,221	237	78× ⚠ Below 100× threshold	
FAT1	chr4:186,617,688-186,621,795	4,108	119× ✓ Covered	
FAT1	chr4:186,628,134-186,628,384	251	329× ✓ Covered	
FAT1	chr4:186,628,468-186,628,783	316	381× ✓ Covered	
FAT1	chr4:186,633,664-186,633,843	180	77× ⚠ Below 100× threshold	
FAT1	chr4:186,636,005-186,636,255	251	328× ✓ Covered	
FAT1	chr4:186,636,565-186,636,934	370	202× ✓ Covered	
FAT1	chr4:186,639,702-186,639,803	102	809× ✓ Covered	
FAT1	chr4:186,663,279-186,663,633	355	34× ⚠ Below 100× threshold	
FAT1	chr4:186,706,543-186,709,847	3,305	107× ✓ Covered	
TERT				
TERT	chr5:1,295,050-1,295,250	201	447× ✓ Covered	
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	407× ✓ Covered	

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
PIK3R1	chr5:68,295,128-68,295,344	217	72x	⚠ Below 100x threshold
PIK3R1	chr5:68,295,400-68,295,508	109	47x	⚠ Below 100x threshold
NSD1				
NSD1	chr5:177,135,084-177,136,050	967	107x	✓ Covered
NSD1	chr5:177,191,864-177,192,039	176	105x	✓ Covered
NSD1	chr5:177,204,100-177,204,312	213	811x	✓ Covered
NSD1	chr5:177,209,616-177,212,215	2,600	1,415x	✓ Covered
NSD1	chr5:177,235,801-177,235,965	165	42x	⚠ Below 100x threshold
NSD1	chr5:177,238,217-177,238,527	311	180x	✓ Covered
NSD1	chr5:177,239,736-177,239,885	150	638x	✓ Covered
NSD1	chr5:177,244,175-177,244,290	116	25x	⚠ Below 100x threshold
NSD1	chr5:177,246,658-177,246,816	159	123x	✓ Covered
NSD1	chr5:177,248,161-177,248,344	184	146x	✓ Covered
NSD1	chr5:177,251,710-177,251,873	164	238x	✓ Covered
NSD1	chr5:177,256,931-177,257,171	241	270x	✓ Covered
NSD1	chr5:177,259,969-177,260,188	220	1,215x	✓ Covered
NSD1	chr5:177,267,542-177,267,738	197	362x	✓ Covered
NSD1	chr5:177,269,582-177,269,827	246	96x	⚠ Below 100x threshold
NSD1	chr5:177,273,652-177,273,804	153	798x	✓ Covered
NSD1	chr5:177,280,545-177,280,854	310	37x	⚠ Below 100x threshold
NSD1	chr5:177,282,445-177,282,601	157	728x	✓ Covered
NSD1	chr5:177,283,767-177,283,948	182	113x	✓ Covered
NSD1	chr5:177,288,799-177,288,945	147	74x	⚠ Below 100x threshold
NSD1	chr5:177,291,934-177,292,178	245	767x	✓ Covered
NSD1	chr5:177,293,812-177,295,476	1,665	165x	✓ Covered
EGFR				
EGFR	chr7:55,173,901-55,174,063	163	103x	✓ Covered
EGFR	chr7:55,174,702-55,174,840	139	52x	⚠ Below 100x threshold
EGFR	chr7:55,181,273-55,181,498	226	232x	✓ Covered
EGFR	chr7:55,191,699-55,191,894	196	6x	⚠ Below 100x threshold
CDKN2A				
CDKN2A	chr9:21,968,212-21,968,262	51	450x	✓ Covered
CDKN2A	chr9:21,970,882-21,971,228	347	163x	✓ Covered
CDKN2A	chr9:21,974,658-21,974,846	189	69x	⚠ Below 100x threshold
CDKN2A	chr9:21,974,658-21,974,847	190	28x	⚠ Below 100x threshold
NOTCH1				
NOTCH1	chr9:136,496,054-136,497,578	1,525	81x	⚠ Below 100x threshold
NOTCH1	chr9:136,498,879-136,499,016	138	361x	✓ Covered
NOTCH1	chr9:136,499,092-136,499,279	188	162x	✓ Covered
NOTCH1	chr9:136,500,532-136,500,867	336	257x	✓ Covered
NOTCH1	chr9:136,501,728-136,501,933	206	100x	✓ Covered
NOTCH1	chr9:136,501,981-136,502,108	128	105x	✓ Covered
NOTCH1	chr9:136,502,252-136,502,508	257	170x	✓ Covered
NOTCH1	chr9:136,503,162-136,503,350	189	441x	✓ Covered
NOTCH1	chr9:136,504,653-136,505,124	472	285x	✓ Covered
NOTCH1	chr9:136,505,290-136,505,901	612	896x	✓ Covered
NOTCH1	chr9:136,506,507-136,506,659	153	849x	✓ Covered
NOTCH1	chr9:136,506,696-136,506,993	298	139x	✓ Covered
NOTCH1	chr9:136,507,285-136,507,457	173	299x	✓ Covered
NOTCH1	chr9:136,507,935-136,508,159	225	79x	⚠ Below 100x threshold
NOTCH1	chr9:136,508,212-136,508,405	194	153x	✓ Covered

FULL PANEL COVERAGE REPORT — Morrison, Toni E. (NGS-2026-052106)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	74×	⚠ Below 100× threshold
NOTCH1	chr9:136,509,713-136,509,981	269	63×	⚠ Below 100× threshold
NOTCH1	chr9:136,510,633-136,510,825	193	36×	⚠ Below 100× threshold
NOTCH1	chr9:136,511,132-136,511,291	160	3,252×	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	48×	⚠ Below 100× threshold
NOTCH1	chr9:136,513,372-136,513,557	186	86×	⚠ Below 100× threshold
NOTCH1	chr9:136,514,490-136,514,722	233	85×	⚠ Below 100× threshold
NOTCH1	chr9:136,515,270-136,515,420	151	2×	⚠ Below 100× threshold
NOTCH1	chr9:136,515,463-136,515,736	274	411×	✓ Covered
NOTCH1	chr9:136,515,961-136,516,114	154	92×	⚠ Below 100× threshold
NOTCH1	chr9:136,517,252-136,517,405	154	56×	⚠ Below 100× threshold
NOTCH1	chr9:136,517,732-136,517,957	226	171×	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	63×	⚠ Below 100× threshold
NOTCH1	chr9:136,518,571-136,518,844	274	57×	⚠ Below 100× threshold
NOTCH1	chr9:136,519,423-136,519,585	163	53×	⚠ Below 100× threshold
NOTCH1	chr9:136,522,830-136,523,208	379	223×	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	198×	✓ Covered
NOTCH1	chr9:136,544,004-136,544,122	119	1,049×	✓ Covered
NOTCH1	chr9:136,545,706-136,545,806	101	214×	✓ Covered

PATIENT & SPECIMEN INFORMATION

Patient name	Austen, Jane C.	Accession	NGS-2026-052104
Date of birth	1967-12-04 (Age 58)	Specimen	Oral swab, soft palate
MRN	40678954	Collected	2026-05-19
Ordering physician	Dr. Wisdom, MD	Received	2026-05-19
Clinical indication	Oral lesion — soft palate	Reported	2026-05-20

RESULT: VARIANTS DETECTED

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
EGFR	NM_005228.5:c.2573T>G	p.Leu858Arg	0.57%	525×	I	Pathogenic
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.30%	669×	I	Pathogenic

VARIANT ANNOTATION

EGFR p.Leu858Arg | Pathogenic (Tier I)

EGFR p.Leu858Arg (L858R) is a well-characterised activating mutation in exon 21 of the EGFR kinase domain. It disrupts the autoinhibitory conformation, leading to constitutive EGFR kinase activation and downstream RAS/MAPK and PI3K/AKT signalling. While canonical in non-small cell lung adenocarcinoma (where it predicts response to EGFR TKIs such as osimertinib), EGFR L858R has been reported in 1–3% of OSCC. Cetuximab (anti-EGFR monoclonal antibody) is FDA-approved in HNSCC and may have enhanced activity in the setting of EGFR activating mutations. Multidisciplinary tumour board review for consideration of EGFR-directed therapy or clinical trial enrolment is recommended. Co-occurring TP53 R175H (see below) may affect response to EGFR-directed therapy and warrants combined consideration.

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His — see interpretation above (Dickens, Charles J., NGS-2026-052103). Co-occurrence of EGFR L858R and TP53 R175H has been associated with resistance to EGFR-directed therapies in other tumour types; multidisciplinary tumour board review is recommended. NOTE: Observed VAF (0.30%) is above the 0.2% reporting threshold and is supported by 2 consensus reads at 669× consensus depth. Orthogonal confirmation is recommended before clinical action.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.2% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.2% with validated sensitivity of 98.4% at ≥20,000× raw input depth (≥400× consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.2% VAF reporting threshold may be present but are not reported. At 2,000× raw sequencing depth, approximately 15–20% of targets may fall below 100× consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

HOTSPOT SCREENING SUMMARY

Patient:

Austen, Jane C.

Accession:

NGS-2026-052104

Hotspots listed below were explicitly interrogated in addition to all variants above 0.2% VAF with ≥2 consensus reads across all 251 targets. Coverage criteria (≥100× consensus depth) were met at all targets.

EGFR			Cov ✓
L858R	c.2573T>G	● DETECTED	
T790M	c.2369C>T	Not detected	
Exon 19 del	c.2235_2249del	Not detected	

KRAS			Cov ✓
G12D	c.35G>T	Not detected	
G12V	c.35G>A	Not detected	
G12C	c.34G>T	Not detected	
G13D	c.38G>A	Not detected	

PIK3CA			Cov ✓
E542K	c.1624G>A	Not detected	
E545K	c.1633G>A	Not detected	
H1047R	c.3140A>G	Not detected	

TERT			Cov ✓
C228T	c.-124C>T	Not detected	
C250T	c.-146C>T	Not detected	

HRAS			Cov ✓
G12V	c.35G>T	Not detected	
G12D	c.35G>A	Not detected	
Q61L	c.182A>T	Not detected	
Q61R	c.182A>G	Not detected	

NOTCH1			Cov ✓
P2415L	c.7244C>T	Not detected	
R1699W	c.5095C>T	Not detected	

TP53			Cov ✓
R175H	c.524G>A	● DETECTED	
R248Q	c.743G>A	Not detected	
R248W	c.742C>T	Not detected	
R273H	c.818G>A	Not detected	
R282W	c.844C>T	Not detected	

REMAINING PANEL GENES — SEQUENCE-LEVEL DETECTION, NO PREDEFINED HOTSPOT LIST

AJUBA · CASP8 · CDKN2A · CUL3 · DPYD · EP300 · FAT1 · FBXW7 · KEAP1 · KMT2D · NFE2L2 · NSD1 · PIK3R1 · PTEN

All variants above 0.2% VAF with ≥2 consensus reads are reported for these genes. No variants meeting reporting criteria were detected. All targets in these genes met minimum coverage criteria (≥100× consensus depth).

Reported by: Dr. Wisdom, MD PhD, FCAP

Report date: 2026-05-20

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)

251 targets, 77,417 bp total. Mean consensus depth: 405x. Targets below 100x consensus: 57 / 251 (23%). Depths are derived from the consensus BAM (UMI deduplication, min-reads ≥ 2). Positions reported 1-based.

Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
DPYD				
DPYD	chr1:97,082,291-97,082,491	201	277x	✓ Covered
DPYD	chr1:97,449,958-97,450,158	201	502x	✓ Covered
DPYD	chr1:97,515,687-97,515,887	201	82x	⚠ Below 100x threshold
DPYD	chr1:97,573,763-97,573,963	201	439x	✓ Covered
PTEN				
PTEN	chr10:87,864,450-87,864,568	119	158x	✓ Covered
PTEN	chr10:87,894,005-87,894,129	125	189x	✓ Covered
PTEN	chr10:87,925,493-87,925,577	85	33x	⚠ Below 100x threshold
PTEN	chr10:87,931,026-87,931,109	84	848x	✓ Covered
PTEN	chr10:87,932,993-87,933,271	279	249x	✓ Covered
PTEN	chr10:87,952,098-87,952,279	182	285x	✓ Covered
PTEN	chr10:87,957,833-87,958,039	207	64x	⚠ Below 100x threshold
PTEN	chr10:87,960,874-87,961,138	265	250x	✓ Covered
PTEN	chr10:87,965,267-87,965,489	223	621x	✓ Covered
HRAS				
HRAS	chr11:532,619-532,775	157	67x	⚠ Below 100x threshold
HRAS	chr11:533,433-533,632	200	28x	⚠ Below 100x threshold
HRAS	chr11:533,746-533,964	219	326x	✓ Covered
HRAS	chr11:534,192-534,342	151	330x	✓ Covered
KRAS				
KRAS	chr12:25,209,778-25,209,931	154	119x	✓ Covered
KRAS	chr12:25,225,594-25,225,793	200	332x	✓ Covered
KRAS	chr12:25,227,214-25,227,432	219	610x	✓ Covered
KRAS	chr12:25,245,254-25,245,404	151	183x	✓ Covered
KMT2D				
KMT2D	chr12:49,021,763-49,021,892	130	433x	✓ Covered
KMT2D	chr12:49,022,023-49,022,171	149	102x	✓ Covered
KMT2D	chr12:49,022,260-49,022,373	114	18x	⚠ Below 100x threshold
KMT2D	chr12:49,022,570-49,022,895	326	153x	✓ Covered
KMT2D	chr12:49,024,558-49,024,728	171	93x	⚠ Below 100x threshold
KMT2D	chr12:49,024,790-49,024,966	177	343x	✓ Covered
KMT2D	chr12:49,026,162-49,027,342	1,181	252x	✓ Covered
KMT2D	chr12:49,027,783-49,027,950	168	109x	✓ Covered
KMT2D	chr12:49,027,989-49,028,161	173	196x	✓ Covered
KMT2D	chr12:49,028,808-49,028,978	171	1,640x	✓ Covered
KMT2D	chr12:49,029,041-49,029,256	216	75x	⚠ Below 100x threshold
KMT2D	chr12:49,029,381-49,029,496	116	145x	✓ Covered
KMT2D	chr12:49,030,260-49,030,459	200	503x	✓ Covered
KMT2D	chr12:49,030,581-49,030,788	208	1,497x	✓ Covered
KMT2D	chr12:49,030,873-49,031,053	181	835x	✓ Covered
KMT2D	chr12:49,031,155-49,033,984	2,830	142x	✓ Covered
KMT2D	chr12:49,034,047-49,034,319	273	125x	✓ Covered
KMT2D	chr12:49,034,390-49,034,496	107	123x	✓ Covered
KMT2D	chr12:49,034,562-49,034,686	125	484x	✓ Covered
KMT2D	chr12:49,034,792-49,034,955	164	227x	✓ Covered
KMT2D	chr12:49,037,105-49,039,009	1,905	259x	✓ Covered
KMT2D	chr12:49,039,202-49,039,378	177	1,415x	✓ Covered
KMT2D	chr12:49,039,415-49,039,637	223	173x	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)					
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status	
KMT2D	chr12:49,039,704–49,041,555	1,852	168×	✓ Covered	
KMT2D	chr12:49,041,635–49,041,725	91	118×	✓ Covered	
KMT2D	chr12:49,041,897–49,042,010	114	253×	✓ Covered	
KMT2D	chr12:49,042,069–49,042,350	282	101×	✓ Covered	
KMT2D	chr12:49,042,541–49,042,665	125	206×	✓ Covered	
KMT2D	chr12:49,042,721–49,042,898	178	87×	⚠ Below 100× threshold	
KMT2D	chr12:49,043,056–49,043,206	151	122×	✓ Covered	
KMT2D	chr12:49,043,343–49,043,448	106	143×	✓ Covered	
KMT2D	chr12:49,043,615–49,043,802	188	50×	⚠ Below 100× threshold	
KMT2D	chr12:49,043,848–49,044,018	171	154×	✓ Covered	
KMT2D	chr12:49,044,180–49,044,324	145	2,165×	✓ Covered	
KMT2D	chr12:49,044,383–49,044,542	160	36×	⚠ Below 100× threshold	
KMT2D	chr12:49,044,724–49,044,985	262	126×	✓ Covered	
KMT2D	chr12:49,045,900–49,045,987	88	169×	✓ Covered	
KMT2D	chr12:49,046,045–49,046,194	150	292×	✓ Covered	
KMT2D	chr12:49,046,240–49,046,444	205	100×	✓ Covered	
KMT2D	chr12:49,046,589–49,046,810	222	86×	⚠ Below 100× threshold	
KMT2D	chr12:49,047,945–49,048,089	145	152×	✓ Covered	
KMT2D	chr12:49,048,639–49,048,789	151	224×	✓ Covered	
KMT2D	chr12:49,049,085–49,049,238	154	325×	✓ Covered	
KMT2D	chr12:49,049,662–49,050,810	1,149	545×	✓ Covered	
KMT2D	chr12:49,050,866–49,052,444	1,579	86×	⚠ Below 100× threshold	
KMT2D	chr12:49,052,544–49,052,729	186	67×	⚠ Below 100× threshold	
KMT2D	chr12:49,052,895–49,053,092	198	211×	✓ Covered	
KMT2D	chr12:49,053,187–49,053,341	155	63×	⚠ Below 100× threshold	
KMT2D	chr12:49,053,456–49,053,661	206	615×	✓ Covered	
KMT2D	chr12:49,053,958–49,054,160	203	154×	✓ Covered	
KMT2D	chr12:49,054,287–49,054,436	150	50×	⚠ Below 100× threshold	
KMT2D	chr12:49,054,508–49,054,771	264	113×	✓ Covered	
KMT2D	chr12:49,054,880–49,055,046	167	231×	✓ Covered	
KMT2D	chr12:49,055,256–49,055,344	89	145×	✓ Covered	
AJUBA					
AJUBA	chr14:22,973,426–22,973,588	163	5,579×	✓ Covered	
AJUBA	chr14:22,974,027–22,974,135	109	88×	⚠ Below 100× threshold	
AJUBA	chr14:22,974,819–22,974,910	92	314×	✓ Covered	
AJUBA	chr14:22,974,954–22,975,124	171	278×	✓ Covered	
AJUBA	chr14:22,976,436–22,976,538	103	108×	✓ Covered	
AJUBA	chr14:22,976,625–22,976,732	108	246×	✓ Covered	
AJUBA	chr14:22,978,324–22,978,465	142	347×	✓ Covered	
AJUBA	chr14:22,981,241–22,982,286	1,046	70×	⚠ Below 100× threshold	
TP53					
TP53	chr17:7,669,592–7,669,710	119	103×	✓ Covered	
TP53	chr17:7,670,589–7,670,735	147	164×	✓ Covered	
TP53	chr17:7,673,515–7,673,628	114	2,014×	✓ Covered	
TP53	chr17:7,673,681–7,673,857	177	753×	✓ Covered	
TP53	chr17:7,674,161–7,674,310	150	423×	✓ Covered	
TP53	chr17:7,674,839–7,674,991	153	176×	✓ Covered	
TP53	chr17:7,675,033–7,675,256	224	144×	🚫 VARIANT DETECTED	
TP53	chr17:7,675,974–7,676,292	319	186×	✓ Covered	
TP53	chr17:7,676,362–7,676,423	62	347×	✓ Covered	
TP53	chr17:7,676,501–7,676,614	114	1,633×	✓ Covered	

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
KEAP1				
KEAP1	chr19:10,486,635–10,486,838	204	923x	✓ Covered
KEAP1	chr19:10,489,172–10,489,388	217	251x	✓ Covered
KEAP1	chr19:10,489,628–10,489,873	246	306x	✓ Covered
KEAP1	chr19:10,491,557–10,492,282	726	30x	⚠ Below 100x threshold
KEAP1	chr19:10,499,375–10,500,053	679	245x	✓ Covered
NFE2L2				
NFE2L2	chr2:177,233,985–177,234,291	307	38x	⚠ Below 100x threshold
CASP8				
CASP8	chr2:201,266,467–201,266,811	345	70x	⚠ Below 100x threshold
CASP8	chr2:201,271,496–201,271,641	146	43x	⚠ Below 100x threshold
CASP8	chr2:201,272,618–201,272,796	179	155x	✓ Covered
CASP8	chr2:201,272,878–201,272,962	85	197x	✓ Covered
CASP8	chr2:201,274,869–201,274,973	105	546x	✓ Covered
CASP8	chr2:201,276,807–201,276,988	182	171x	✓ Covered
CASP8	chr2:201,284,796–201,285,337	542	239x	✓ Covered
CASP8	chr2:201,286,439–201,286,611	173	94x	⚠ Below 100x threshold
CUL3				
CUL3	chr2:224,474,228–224,474,396	169	165x	✓ Covered
CUL3	chr2:224,478,180–224,478,365	186	104x	✓ Covered
CUL3	chr2:224,481,872–224,482,098	227	274x	✓ Covered
CUL3	chr2:224,495,812–224,495,986	175	88x	⚠ Below 100x threshold
CUL3	chr2:224,497,733–224,497,869	137	630x	✓ Covered
CUL3	chr2:224,500,343–224,500,507	165	228x	✓ Covered
CUL3	chr2:224,502,945–224,503,092	148	132x	✓ Covered
CUL3	chr2:224,503,632–224,503,842	211	135x	✓ Covered
CUL3	chr2:224,505,936–224,506,152	217	820x	✓ Covered
CUL3	chr2:224,506,838–224,507,023	186	241x	✓ Covered
CUL3	chr2:224,511,334–224,511,602	269	512x	✓ Covered
CUL3	chr2:224,513,504–224,513,658	155	756x	✓ Covered
CUL3	chr2:224,514,592–224,514,792	201	586x	✓ Covered
CUL3	chr2:224,535,508–224,535,661	154	123x	✓ Covered
CUL3	chr2:224,557,639–224,557,876	238	666x	✓ Covered
CUL3	chr2:224,584,924–224,585,029	106	136x	✓ Covered
EP300				
EP300	chr22:41,092,985–41,093,118	134	1,271x	✓ Covered
EP300	chr22:41,117,167–41,117,841	675	98x	⚠ Below 100x threshold
EP300	chr22:41,125,844–41,126,060	217	75x	⚠ Below 100x threshold
EP300	chr22:41,127,467–41,127,768	302	44x	⚠ Below 100x threshold
EP300	chr22:41,129,870–41,130,023	154	248x	✓ Covered
EP300	chr22:41,131,368–41,131,653	286	493x	✓ Covered
EP300	chr22:41,135,793–41,135,926	134	187x	✓ Covered
EP300	chr22:41,137,633–41,137,810	178	1,871x	✓ Covered
EP300	chr22:41,140,120–41,140,277	158	201x	✓ Covered
EP300	chr22:41,141,028–41,141,242	215	52x	⚠ Below 100x threshold
EP300	chr22:41,146,719–41,146,836	118	2,457x	✓ Covered
EP300	chr22:41,147,817–41,147,966	150	92x	⚠ Below 100x threshold
EP300	chr22:41,149,018–41,149,195	178	960x	✓ Covered
EP300	chr22:41,149,741–41,150,218	478	77x	⚠ Below 100x threshold
EP300	chr22:41,151,813–41,152,032	220	41x	⚠ Below 100x threshold
EP300	chr22:41,152,186–41,152,370	185	44x	⚠ Below 100x threshold

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
EP300	chr22:41,154,975-41,155,133	159	114×	✓ Covered
EP300	chr22:41,157,149-41,157,428	280	389×	✓ Covered
EP300	chr22:41,158,392-41,158,520	129	843×	✓ Covered
EP300	chr22:41,160,622-41,160,742	121	1,391×	✓ Covered
EP300	chr22:41,162,703-41,162,799	97	390×	✓ Covered
EP300	chr22:41,164,033-41,164,150	118	62×	⚠ Below 100× threshold
EP300	chr22:41,166,579-41,166,686	108	121×	✓ Covered
EP300	chr22:41,168,429-41,168,619	191	26×	⚠ Below 100× threshold
EP300	chr22:41,168,701-41,168,887	187	58×	⚠ Below 100× threshold
EP300	chr22:41,169,483-41,169,636	154	175×	✓ Covered
EP300	chr22:41,170,386-41,170,591	206	152×	✓ Covered
EP300	chr22:41,172,479-41,172,683	205	810×	✓ Covered
EP300	chr22:41,173,603-41,173,804	202	217×	✓ Covered
EP300	chr22:41,176,227-41,176,548	322	77×	⚠ Below 100× threshold
EP300	chr22:41,176,753-41,178,973	2,221	119×	✓ Covered
PIK3CA				
PIK3CA	chr3:179,218,190-179,218,354	165	217×	✓ Covered
PIK3CA	chr3:179,234,074-179,234,381	308	29×	⚠ Below 100× threshold
FBXW7				
FBXW7	chr4:152,325,986-152,326,251	266	168×	✓ Covered
FAT1				
FAT1	chr4:186,588,575-186,589,240	666	54×	⚠ Below 100× threshold
FAT1	chr4:186,595,669-186,595,846	178	1,364×	✓ Covered
FAT1	chr4:186,596,520-186,597,191	672	310×	✓ Covered
FAT1	chr4:186,597,662-186,597,812	151	146×	✓ Covered
FAT1	chr4:186,597,952-186,598,145	194	1,355×	✓ Covered
FAT1	chr4:186,599,878-186,600,380	503	100×	✓ Covered
FAT1	chr4:186,601,249-186,601,446	198	159×	✓ Covered
FAT1	chr4:186,602,883-186,603,054	172	43×	⚠ Below 100× threshold
FAT1	chr4:186,603,156-186,603,997	842	216×	✓ Covered
FAT1	chr4:186,604,357-186,604,594	238	113×	✓ Covered
FAT1	chr4:186,606,050-186,606,233	184	57×	⚠ Below 100× threshold
FAT1	chr4:186,609,163-186,609,340	178	678×	✓ Covered
FAT1	chr4:186,609,781-186,610,035	255	688×	✓ Covered
FAT1	chr4:186,611,366-186,611,795	430	534×	✓ Covered
FAT1	chr4:186,613,089-186,613,362	274	376×	✓ Covered
FAT1	chr4:186,614,171-186,614,364	194	81×	⚠ Below 100× threshold
FAT1	chr4:186,616,985-186,617,221	237	697×	✓ Covered
FAT1	chr4:186,617,688-186,621,795	4,108	82×	⚠ Below 100× threshold
FAT1	chr4:186,628,134-186,628,384	251	89×	⚠ Below 100× threshold
FAT1	chr4:186,628,468-186,628,783	316	311×	✓ Covered
FAT1	chr4:186,633,664-186,633,843	180	106×	✓ Covered
FAT1	chr4:186,636,005-186,636,255	251	132×	✓ Covered
FAT1	chr4:186,636,565-186,636,934	370	67×	⚠ Below 100× threshold
FAT1	chr4:186,639,702-186,639,803	102	293×	✓ Covered
FAT1	chr4:186,663,279-186,663,633	355	202×	✓ Covered
FAT1	chr4:186,706,543-186,709,847	3,305	523×	✓ Covered
TERT				
TERT	chr5:1,295,050-1,295,250	201	738×	✓ Covered
PIK3R1				
PIK3R1	chr5:68,294,516-68,294,698	183	449×	✓ Covered

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)					
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status	
PIK3R1	chr5:68,295,128-68,295,344	217	1,550×	✓	Covered
PIK3R1	chr5:68,295,400-68,295,508	109	921×	✓	Covered
NSD1					
NSD1	chr5:177,135,084-177,136,050	967	684×	✓	Covered
NSD1	chr5:177,191,864-177,192,039	176	82×	⚠	Below 100× threshold
NSD1	chr5:177,204,100-177,204,312	213	341×	✓	Covered
NSD1	chr5:177,209,616-177,212,215	2,600	288×	✓	Covered
NSD1	chr5:177,235,801-177,235,965	165	439×	✓	Covered
NSD1	chr5:177,238,217-177,238,527	311	363×	✓	Covered
NSD1	chr5:177,239,736-177,239,885	150	212×	✓	Covered
NSD1	chr5:177,244,175-177,244,290	116	264×	✓	Covered
NSD1	chr5:177,246,658-177,246,816	159	69×	⚠	Below 100× threshold
NSD1	chr5:177,248,161-177,248,344	184	600×	✓	Covered
NSD1	chr5:177,251,710-177,251,873	164	163×	✓	Covered
NSD1	chr5:177,256,931-177,257,171	241	345×	✓	Covered
NSD1	chr5:177,259,969-177,260,188	220	2,164×	✓	Covered
NSD1	chr5:177,267,542-177,267,738	197	245×	✓	Covered
NSD1	chr5:177,269,582-177,269,827	246	1,381×	✓	Covered
NSD1	chr5:177,273,652-177,273,804	153	421×	✓	Covered
NSD1	chr5:177,280,545-177,280,854	310	143×	✓	Covered
NSD1	chr5:177,282,445-177,282,601	157	346×	✓	Covered
NSD1	chr5:177,283,767-177,283,948	182	477×	✓	Covered
NSD1	chr5:177,288,799-177,288,945	147	279×	✓	Covered
NSD1	chr5:177,291,934-177,292,178	245	361×	✓	Covered
NSD1	chr5:177,293,812-177,295,476	1,665	482×	✓	Covered
EGFR					
EGFR	chr7:55,173,901-55,174,063	163	296×	✓	Covered
EGFR	chr7:55,174,702-55,174,840	139	181×	✓	Covered
EGFR	chr7:55,181,273-55,181,498	226	676×	✓	Covered
EGFR	chr7:55,191,699-55,191,894	196	232×	●	VARIANT DETECTED
CDKN2A					
CDKN2A	chr9:21,968,212-21,968,262	51	484×	✓	Covered
CDKN2A	chr9:21,970,882-21,971,228	347	1,874×	✓	Covered
CDKN2A	chr9:21,974,658-21,974,846	189	730×	✓	Covered
CDKN2A	chr9:21,974,658-21,974,847	190	132×	✓	Covered
NOTCH1					
NOTCH1	chr9:136,496,054-136,497,578	1,525	401×	✓	Covered
NOTCH1	chr9:136,498,879-136,499,016	138	84×	⚠	Below 100× threshold
NOTCH1	chr9:136,499,092-136,499,279	188	61×	⚠	Below 100× threshold
NOTCH1	chr9:136,500,532-136,500,867	336	618×	✓	Covered
NOTCH1	chr9:136,501,728-136,501,933	206	276×	✓	Covered
NOTCH1	chr9:136,501,981-136,502,108	128	507×	✓	Covered
NOTCH1	chr9:136,502,252-136,502,508	257	775×	✓	Covered
NOTCH1	chr9:136,503,162-136,503,350	189	1,836×	✓	Covered
NOTCH1	chr9:136,504,653-136,505,124	472	244×	✓	Covered
NOTCH1	chr9:136,505,290-136,505,901	612	232×	✓	Covered
NOTCH1	chr9:136,506,507-136,506,659	153	96×	⚠	Below 100× threshold
NOTCH1	chr9:136,506,696-136,506,993	298	70×	⚠	Below 100× threshold
NOTCH1	chr9:136,507,285-136,507,457	173	245×	✓	Covered
NOTCH1	chr9:136,507,935-136,508,159	225	90×	⚠	Below 100× threshold
NOTCH1	chr9:136,508,212-136,508,405	194	81×	⚠	Below 100× threshold

FULL PANEL COVERAGE REPORT — Austen, Jane C. (NGS-2026-052104)				
Gene	chr:start-end (1-based)	Len (bp)	Depth	Coverage status
NOTCH1	chr9:136,508,850-136,509,091	242	177x	✓ Covered
NOTCH1	chr9:136,509,713-136,509,981	269	41x	⚠ Below 100x threshold
NOTCH1	chr9:136,510,633-136,510,825	193	717x	✓ Covered
NOTCH1	chr9:136,511,132-136,511,291	160	1,355x	✓ Covered
NOTCH1	chr9:136,513,001-136,513,154	154	5,649x	✓ Covered
NOTCH1	chr9:136,513,372-136,513,557	186	111x	✓ Covered
NOTCH1	chr9:136,514,490-136,514,722	233	161x	✓ Covered
NOTCH1	chr9:136,515,270-136,515,420	151	329x	✓ Covered
NOTCH1	chr9:136,515,463-136,515,736	274	96x	⚠ Below 100x threshold
NOTCH1	chr9:136,515,961-136,516,114	154	455x	✓ Covered
NOTCH1	chr9:136,517,252-136,517,405	154	223x	✓ Covered
NOTCH1	chr9:136,517,732-136,517,957	226	258x	✓ Covered
NOTCH1	chr9:136,518,117-136,518,312	196	31x	⚠ Below 100x threshold
NOTCH1	chr9:136,518,571-136,518,844	274	131x	✓ Covered
NOTCH1	chr9:136,519,423-136,519,585	163	177x	✓ Covered
NOTCH1	chr9:136,522,830-136,523,208	379	635x	✓ Covered
NOTCH1	chr9:136,523,697-136,523,999	303	18x	⚠ Below 100x threshold
NOTCH1	chr9:136,544,004-136,544,122	119	81x	⚠ Below 100x threshold
NOTCH1	chr9:136,545,706-136,545,806	101	183x	✓ Covered